Automated Spray Control using Deep Learning and Image Processing

Victor Azad
victor.azad.20e@iitram.ac.in
IITRAM, Ahmedabad, India

Shreyank Kumar
shreyank.kumar.21m@iitram.ac.in
IITRAM, Ahmedabad, India

Ram Narayan Yadav
ramnarayan@iitram.ac.in
IITRAM, Ahmedabad, India

ABSTRACT

The rapid and accurate identification of plant diseases is a critical challenge in agriculture, impacting cost, crop health and agricultural productivity. Traditional methods of disease detection heavily rely on manual inspection, which can be time-consuming, subjective, and prone to human error. In this work, we propose an end-to-end system for automated detection of disease with severity level in tomato plants using a fusion of deep learning methods with image processing techniques. Farmers can reduce the needless and excessive use of pesticides by using targeted and optimized treatment plans by precisely measuring the severity level of the disease. We have developed a testbed to show the effectiveness of our proposed end-to-end spray control in terms of accuracy of disease classification, severity detection, and amount of pesticide reduction. In comparison to the traditional spray approach, our suggested setup resulted in 20% pesticide reductions and a 29.68% energy savings.

KEYWORDS

Bacterial Spot, Deep Learning, Image Processing, Spray Control

ACM Reference Format:
Victor Azad, Shreyank Kumar, and Ram Narayan Yadav. 2023. Automated Spray Control using Deep Learning and Image Processing. In *The 29th Annual International Conference on Mobile Computing and Networking (ACM MobiCom '23), October 2–6, 2023, Madrid, Spain.* ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/3570361.3615757

ACM MobiCom '23, October 2–6, 2023, Madrid, Spain

ACM ISBN 978-1-4503-9990-6/23/10…$15.00
https://doi.org/10.1145/3570361.3615757

1 INTRODUCTION

The importance of plants in food production cannot be overstated, as they form the foundation of our agricultural systems and provide the majority of our sustenance. However, disease in plants poses a significant threat to food production, causing substantial losses in crop yield and quality. Thus, plant disease detection is of paramount importance in agriculture as it enables early identification and targeted management of diseases, reducing crop losses and ensuring food security. Accurate disease detection also aids in the adoption of appropriate control measures, minimizing the need for excessive pesticide use and promoting sustainable agricultural practices. However, manually detecting disease in plants is very laborious and time-consuming work. The use of machine learning and computer vision techniques has gained significant attention in the detection of diseases in plants. These methods offer the potential for automated and accurate identification of plant diseases, enabling early intervention and effective disease management.

Numerous studies have focused on developing accurate and efficient methods for identifying the presence or absence of diseases in plants based on visual symptoms and patterns[2][4]. However, there is a noticeable gap in the literature when it comes to estimating the severity or intensity of the disease. Estimating the severity of a disease is crucial for effective disease management and decision-making in agricultural practices. In this work, we propose an end-to-end system that combines deep learning and image processing techniques to address the challenge of both disease detection and severity estimation in tomato plants as shown in Figure 1.

2 METHODOLOGY

In this section, we discuss the dataset that we have used for our study. We also discuss the proposed end-to-end architecture for automated spray control system for disease management in agriculture.

2.1 Description of the dataset

In this work, we use the Plant Village dataset that consists of 18162 images and segmented binary mask images [3]. This dataset serves as a valuable resource for developing

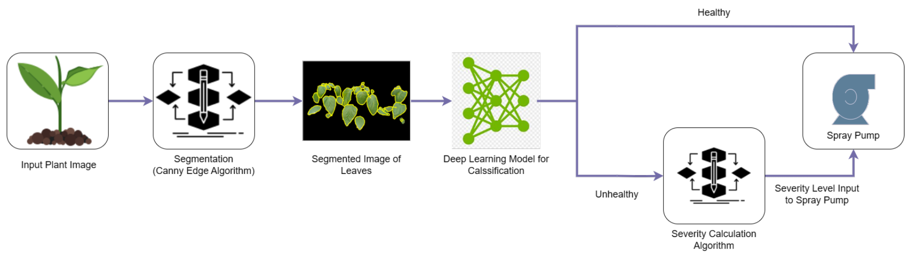

Figure 1: End-to-end system architecture

and evaluating machine learning algorithms and computer vision techniques aimed at automated disease diagnosis in tomato plants. This dataset was additionally developed for tomato leaves and it comprises 10 classes, with one healthy images class and nine unhealthy images classes. The nine unhealthy classes were also divided into five subcategories, including microscopic organisms, infections, growths, molds, and parasites.

2.2 End-to-end system architecture

By leveraging the power of deep learning models, such as vision transformer [1], we aim to automatically detect disease symptoms and classify them into specific disease categories. Additionally, through advanced image processing algorithms, we seek to quantify the severity of the disease based on the extent of damage observed in the plant tissues as shown in Figure 2. The proposed system offers a comprehensive approach to enable farmers and researchers to obtain more detailed information about the severity of the disease in tomato plants. By providing a quantitative measure of disease severity, it enhances the decision-making process and allows for more targeted and effective disease management strategies. Ultimately, the combination of deep learning and image processing methods in this system holds great potential for advancing the field of plant disease diagnosis and management, contributing an automatic spray to improve cost in terms of pesticide uses and power consumption.

3 RESULTS AND EXPERIMENTAL SETUP

We tested our developed algorithm on system-generated data. The setup is created to examine the effectiveness of the proposed severity algorithm in terms of real-time pesticide conservation. The entire setup consists of an Arduino Uno, a motor driver (L293d), a DC water pump (Spray Pump), a DC power source, an inlet container, an output container, and a motor driver as shown in Figure 4. In the setup, we first determined the severity level of the leaf using the proposed severity estimate algorithm, and we then recorded the severity levels in a CSV file. Then, as indicated in Table 2, we delivered voltage to the motor pump in accordance with those severity levels from the CSV data. As spray pumps are typically supplied by constant voltage supplies, we started by running the pump at full power supply. then by running the machine for 17 seconds, 0.5 seconds for each sample, they were able to determine how much pesticide was applied to the 34 samples.

Parameter	Pesticide	Energy
Standard Mechanism	4 ltr	96 Joules
Proposed Mechanism	3.2 ltr	67.5 Joules
% Improvement	20%	29.6875%

Table 1: Improvement in amount of pest and energy consumption using proposed method

3.1 Tomato disease detection

We used modified vision transformer with DeiT model based distillation token concept for tomato disease detection, model performed very well and detected the disease in tomato plants with 99.31% accuracy. The confusion matrix for the modified vision transformer is shown in Figure 3.

3.2 Pesticide conservation

The conventional technique of spraying, which involves operating a pump at a steady supply, is used as a reference for comparison with our proposed adaptive approach. We operated the pump for 17 seconds. Next, we evaluated the pesticide application using our proposed method, which involves regulating the pump's power supply in line with the severity levels. Table 1 shows the percentage of improvement. This experiment demonstrates that, compared to the Constant spraying mechanism, our proposed system uses less amount of pesticide. So the proposed system can be highly cost-effective for large-scale deployment.

3.3 Energy conservation

We compare the energy consumption of our proposed end-to-end system with the conventional technique (running the spray pump at constant speed). From Table 1, there is

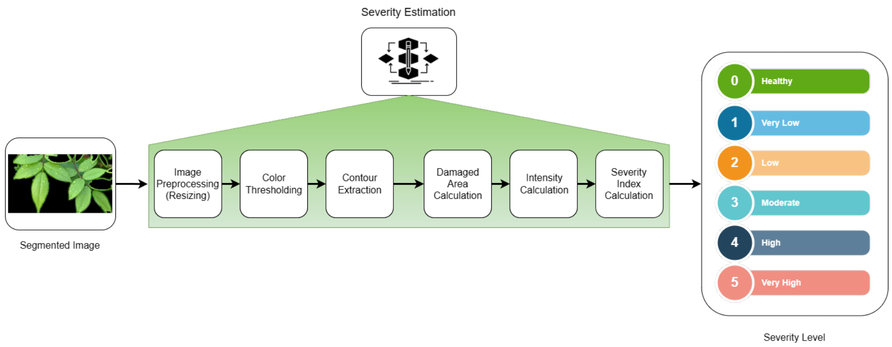

Figure 2: Proposed severity estimation algorithm

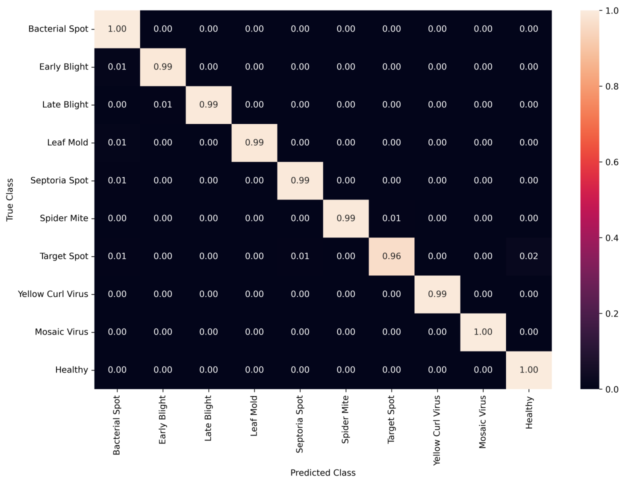

Figure 3: Confusion matrix of the tomato disease detection results using modified vision transformer

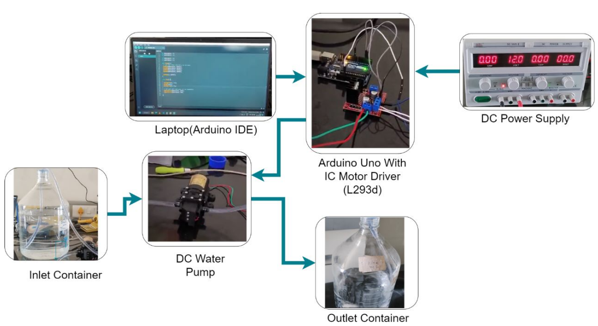

Figure 4: Experimental setup for automatic spray control

a significant improvement in power consumption with our proposed end-to-end system. This experiment shows that our proposed method uses less energy for spraying the same amount of agricultural field that the conventional process. So, the proposed algorithm can be deployed on agriculture drones for spray control.

4 CONCLUSION

In this work, we propose an end-to-end system for automated detection of disease with severity level in tomato plants using a deep learning method (for disease classification) with image processing techniques (severity estimation). We have developed a testbed to show the effectiveness of our proposed end-to-end spray control in terms of accuracy of disease classification, severity detection, and amount of pesticide reduction and energy consumption. In the future, we aim to use deep learning-based techniques to enhance the image segmentation component. Furthermore, the integration of hyper-spectral imaging with conventional computer vision techniques represents an exciting avenue for future exploration of this study. By fusing hyper-spectral data with computer vision algorithms, we will potentially enhance the accuracy of severity estimation.

ACKNOWLEDGMENTS

This work was supported by TIH-IoT under the TIH-IoT CHANAKYA Fellowship Program.

REFERENCES

[1] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).
[2] A. Fuentes, S. Yoon, and D. S. Park. 2018. Tomato diseases classification using deep convolutional neural networks. *Computers and Electronics in Agriculture* 153 (2018), 46–56.
[3] David Hughes, Marcel Salathé, et al. 2015. An open access repository of images on plant health to enable the development of mobile disease diagnostics. *arXiv preprint arXiv:1511.08060* (2015).
[4] Jui-Feng Yeh, Kuei-Mei Lin, Chen-Yu Lin, and Jen-Chun Kang. 2023. Intelligent Mango Fruit Grade Classification Using AlexNet-SPP With Mask R-CNN-Based Segmentation Algorithm. *IEEE Transactions on AgriFood Electronics* (2023).